JAIDEV EDUCATION SOCIETY'S

JD COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Electronics and Telecommunication Engineering

"Rectifying Ideas, Amplifying Knowledge"

Session 2023-24

VISION	MISSION
"To be a Department providing high quality & globally competent knowledge of concurrent technologies in the field of Electronics and Telecommunication."	1. To provide quality teaching learning process through well-developed educational environment and dedicated faculties. 2. To produce competent technocrats of high standards satisfying the needs of all stakeholders.

Program: B.Tech in Electronics and Telecommunication Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III		E- Waste Management	3	-	-	3

Prerequisites for the course	
1	Basic knowledge of environmental science, engineering, chemistry, or a related field is beneficial.

Prior Reading Material/ useful links	
1	https://nptel.ac.in/courses/105105169
2	https://www.udemy.com/course/e-waste-management/
3	https://iimm.org/e-waste-management/

Course Objectives:

Sr.No	Course Objectives
1	To enable students to understand scenario of E-waste.
2	To discuss key elements of E-waste management.
3	To impart life skills about E waste management in routine daily life to minimize the hazards.
4	To assess the need for stricter environmental regulations in India, including Extended Producer Responsibility (EPR) programs, import controls, and international legislation like the Basel Convention and the WEEE Directive.

CourseOutcomes: At the end of the course, students will be able to:

Sr.No	Course Outcome Number	CO statement
1	CO1	Understand the fundamental concepts and current issues related to e-waste management.
2	CO2	Apply various principles and practices for e-waste management.
3	CO3	Analyze the environmental and human health impacts of e-waste and evaluate potential solutions.
4	CO4	Evaluate the effectiveness of existing national and international regulations for e-waste management.
5	CO5	Create an informed and responsible plan for e-waste disposal in a specific context.

Syllabus:

Course Contents	
UnitI	**Introduction:**E- waste: composition and generation. Global context in e- waste; E-waste pollutants, E waste hazardous properties, Effects of pollutant (E- waste) on human health and surrounding environment, domestic e-waste disposal, Basic principles of E waste management, Component of E waste management, Technologies for recovery of resources from electronic waste, resource recovery potential of e-waste, steps in recycling and recovery of materials-mechanical processing, technologies for recovery of materials, occupational and environmental health perspectives of recycling e-waste in India**.** **[8Hrs]**
UnitII	**E-waste hazardous on Global trade**: Essential factors in global waste trade economy, Waste trading as a quint essential part of electronic recycling, Free trade agreements as a means ofwaste trading. Import of hazardous e-waste in India; India's stand on liberalizing import rules, E-waste economy in the organized and unorganized sector. Estimation and recycling of e-waste in metro cities of India. **[8Hrs]**
UnitIII	**E-waste control measures:** Need for stringent health safeguards and environmental protection laws in India, Extended Producers Responsibility (EPR), Import of e-waste permissions, Producer-Public-Government cooperation, Administrative Controls & Engineering controls, monitoring of compliance of Rules, Effective regulatory mechanism strengthened by manpower and technical expertise, Reduction of waste at source. **[8 Hrs]**
Unit IV	**E-waste (Management and Handling) Rules, 2011:** E-Waste (Management) Rules, 2016 - Salient Features and its likely implication. Government assistance for TSDFs. **[8Hrs]**
Unit V	**The International Legislation:** The Basel Convention; The Bamako Convention. The Rotterdam Convention. Waste Electrical and Electronic Equipment (WEEE) Directive in the European Union, Restrictions of Hazardous Substances (RoHS) Directive.**[8Hrs]**
Text Books	
1	E-waste: implications, regulations, and management in India and current global best practices, Johri R., TERI Press, New Delhi
2	Electronic Waste ManagementScience, Hester R.E., and Harrison R.M. 2009
Reference Books	
1	Introduction to E-Waste Management, Laxmi Raghupathy,TERI Press, New Delhi
2	Fowler B, Electronic Waste – 1st Edition (Toxicology and Public Health Issues), 2017,Elsevier
3	Gaidajis, G., Angelakoglou, K.and Aktsoglou, D., 2010. E-waste: environmental problems and current management. Journal of Engineering Science and Technology Review, 3(1), pp.193-199
Useful links	
1	https://news.mit.edu/2013/ewaste-mit
2	https://archive.nptel.ac.in/courses/105/105/105105169/
3	https://www.arsdcollege.ac.in/wp-content/uploads/NEP-SEC-E-waste_sustainability.pdf

Secretary	Chairman	Chairman
BOS	**BOS**	**Acd.Council**